

MERCHANT SERVICES · DEEP DIVE

Testing payment decisions the way we test software

Methodology, worked example and full requirements. Companion to the 15-slide Decision Platform executive summary — read that first; this deck is here for the questions it provokes.

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Every figure in this deck is ILLUSTRATIVE (a worked example) or CALCULATED (derived from stated inputs). Nothing here is ACTUAL — no experiment has run. That is what the ask is for.

Three costs we pay on every decision change

Slow

A routing or credential change is a module release. Weeks to quarters — and the release re-tests everything else in the module, so the cost is not proportional to the change.

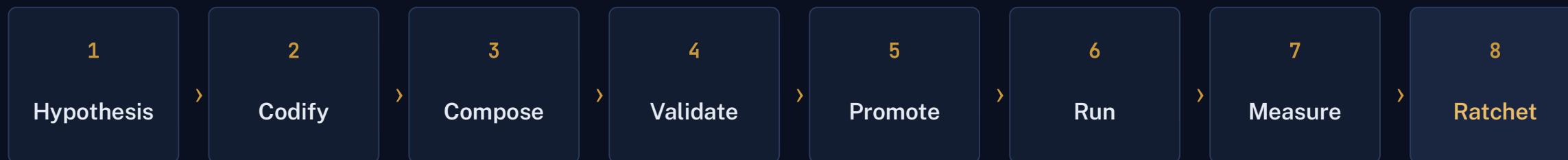
Unprovable

We compare before and after. That confounds the change with seasonality, mix shift, issuer behaviour and every other change shipped the same month. We cannot say what anything was worth.

Unrepeatable

No counterfactual is recorded, so there is nothing to train a model on and nothing to evaluate one against. Every model proposal starts from zero evidence.

One loop, eight steps, days not quarters



every run leaves labelled, counterfactual data behind

The unit of change moves from the module version to the decision.

A strategy is a registered artefact bound to a decision point, not code in a release. That single shift is what makes a change fast to ship, safe to expose, cheap to reverse and — the part that compounds — measurable against a fair baseline.

THREE WORDS

The vocabulary the rest of this deck uses

DECISION POINT

A place in the payment flow where a choice is made.

Which credential to present · whether to authenticate · which processor · what to retry. Five today, extensible.

STRATEGY

How one decision point decides, for one population.

A registered, versioned artefact with an owner, a fallback and an expiry. Deployed without a release.

FLOW

A complete path — one strategy bound at every decision point.

Named and versioned. A variant flow differs from its baseline at exactly one decision point.

A variant flow differs by exactly one strategy. That is the whole design — it is what makes any difference in outcome attributable to that one change, without argument.

Two experiments, one mechanism, opposite directions

Both ask the same question — is the vault network token better than the credential the merchant supplies? — of two different client populations, and expect opposite answers.

EXPERIMENT 1

Does the vault earn its place?

Population	Clients already on the Credential Vault
Control	Vault network token · today
Variant	Merchant-supplied credential · 5%
Expected	Variant performs worse
Proves	What the vault is worth to a client who has it
Used for	Renewal, and defending the line item

EXPERIMENT 2

Is the vault worth adopting?

Population	Clients on BAU passthrough today
Control	Merchant-supplied credential · today
Variant	Vault network token · 1–2%
Expected	Variant performs better
Proves	What the vault would be worth to a prospect
Used for	Adoption, on the client’s own traffic

If the two gaps agree, the vault’s advantage is a property of the credential. If they disagree, it is a property of the client — and knowing which is worth more than either result on its own.

Written before anything is built

EXPERIMENT 1

"For US ecommerce card-on-file transactions on merchants already using the Credential Vault, where the vault holds an active network token and the merchant also supplies its own credential, presenting the merchant-supplied credential instead will reduce transaction approval by at least 25 bps."

REQUIREMENT

WHY

One sentence

Population, treatment, metric, effect size and a guardrail — nothing else

One metric

Transaction approval rate. Not a dashboard of twelve

A number

25 bps. It determines the sample size, which determines the duration

A direction

Stated up front. A test that would be reported as a win either way is not a test

Pre-registered

Recorded before traffic moves, so the result cannot be re-framed after the fact

We already sell the vault on this claim and have never measured it. The direction is not the open question — the size of the gap is, because that is the number a renewal conversation needs.

The strategy is an artefact, not a release

The authorization path, and where the strategy attaches



```
strategy Cred_MerchantSupplied_v1
  decision_point D4.1 credential selection
  class DETERMINISTIC
  population US ecom credit CIT · vault client
    · active NWT AND merchant credential
  definition present the merchant-supplied credential
  fallback Cred_VaultNetworkToken_v4 ← the incumbent
  owner credential.product
  expiry +180d
```

No engineering release.

- Named for what it does, never Strategy1
- Scoped to a population, not to everything
- Names its fallback — never implicit
- Declared as the weaker arm, not discovered to be

One strategy bound at every decision point

FLOW	ROLE	D4.1 CREDENTIAL	D4.2 RETRY	DIFFERS AT
F2_VaultNWT_v6	baseline	Cred_VaultNetworkToken_v4	Attempt_Standard_v1	—
F3_MerchantCred_v1	variant	Cred_MerchantSupplied_v1	Attempt_Standard_v1	D4.1

Every other binding — D1 through D4, D5, D6 — is identical.

Not by convention. The compiler verifies it and refuses to build a variant that differs at two decision points.

Why one difference and not two

Change two things and a 37 bps win tells you nothing about which change produced it — or whether one produced 55 and the other cost 18. If it loses, a good change gets discarded with a bad one. Two changes means two flows on two layers, running at the same time.

Replay it against traffic that already happened

The candidate flow is run against 30 days of logged transactions. Nothing is dispatched. The output is an exact count of decisions that would change, and for whom.

12.8M

transactions replayed

48,291

decisions would change

0.38%

of eligible traffic

0

policy conflicts introduced

This number is exact, not estimated.

Policy and deterministic strategies are reproducible, so replaying them against logged context tells us precisely which transactions would have been decided differently. What it cannot tell us is whether those decisions would have been approved — that is a counterfactual, and it is what the experiment is for. The two are reported separately and never merged.

Available everywhere, carrying nothing

What promotion does

- Registration gates run — around ten, all mechanical
- Flow compiles and is cryptographically signed
- Bundle distributes globally, p99 under 60 seconds
- Flow is resolvable in every region

What promotion does not do

- Move any traffic. Exposure starts at zero
- Change any merchant's behaviour
- Require a deployment window or a release train
- Need to be undone — exposure is a pointer

Promotion and exposure are separate acts.

A flow can sit in production at 0% indefinitely. Traffic moves only when an experiment or a ramp says so, and it stops moving by changing a number — never by a rollback deployment, never by a data migration.

Weights, window, and who qualifies

SETTING	VALUE	WHY IT IS THAT
Control	F2_VaultNWT_v6 97% → 95%	Today's behaviour for a vault client
Variant	F3_MerchantCred_v1 3% → 5%	Deliberately the weaker arm. Kept small
Holdout	2%, never assigned	Measures the platform, not this test
Qualifies	vault client · active NWT AND merchant credential present · US ecom	Both credentials must exist to compare
Randomised on	customer	Not per transaction — issuers notice
Assigned	after the credential gate	Only where the treatment can land
Duration	to 66,900 variant enrolments	Sample size, not a calendar date
Stops early if	gap exceeds -80 bps, or any guardrail	Automatic. Not a meeting

Illustrative configuration. 66,900 variant enrolments detects a 37 bps gap at 95/5 with 80% power — about 1.34M qualifying transactions in total.

BEFORE WE STARTED

Four things that would have invalidated the result

WHAT WE NEARLY DID

WHY IT BREAKS

Changed two decision points in one flow

A win is unattributable to either change

Qualified on the network token alone

Part of the test arm has no merchant credential to present

Randomised per transaction

The same customer flips credential form; issuers read that as risk

Widened the weaker arm to finish sooner

Exploration cost scales with exposure — and it lands on a named merchant

All four were caught by the registration form, not by a reviewer.

Every one is a mechanical check against a required field. That is the point of making them mechanical — the review meeting gets to discuss the hypothesis instead of the statistics.

Exposure, pace and guardrails, in real time

52,340

qualifying today

5.0%

variant share (target 5.0%)

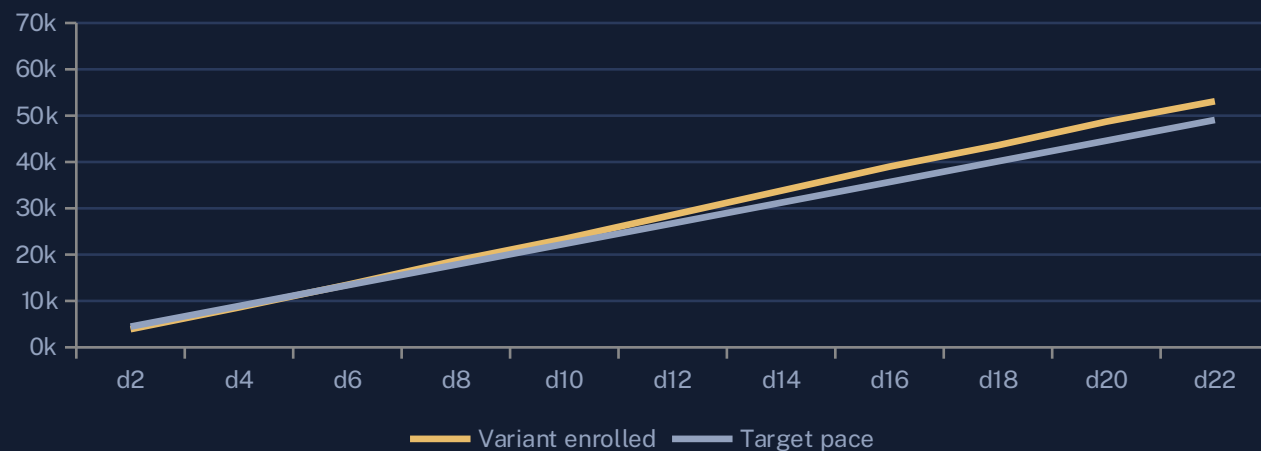
97.6%

treatment fidelity

22 / 30

days to sample size

Enrolment pace — variant arm



Guardrails

Approval gap	-37 bps	●
Abort floor	-80 bps	●
Fraud proxy	+2.2%	●
Dispute proxy	+1.8%	●
Cost of acceptance	-1.4 bps	●
Per-merchant harm	none	●
Latency	no change	●

Illustrative. A breach removes the arm from allocation within one propagation cycle. The abort floor matters here because the variant is knowingly the weaker arm.

Was it in the experiment, and what decided what

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Experiment

In experiment	yes
Experiment	Vault Value · E1
Arm	variant
Flow	F3_MerchantCred_v1
Assigned on	customer cus_C08
Assigned at	after credential gate
Propensity	0.05
Treatment delivered	yes

Decisions applied

POINT	STRATEGY	CHOSE	WOULD HAVE
D4.1	Cred_MerchantSupplied_v1	merchant cred	vault NWT
D4.2	Attempt_Standard_v1	no action	no action
D5	Auth_RiskAdaptive_v3	none	none
D6	Route_Balanced_v7	proc_a	proc_b

"Would have" is what the baseline flow would have decided — not what the issuer would have answered. That is the counterfactual, and it cannot be reconstructed after the fact.

A scorecard, not an approval rate

Variant is the merchant-supplied credential. A negative approval delta means the vault won.

MEASURE	CONTROL · VAULT	VARIANT · MERCHANT	DELTA	INTERVAL
Transaction approval	87.68%	87.31%	-37 bps	[-54, -20] pass
First-attempt approval	84.44%	84.02%	-42 bps	[-60, -24] pass
Cost of acceptance	—	—	-1.4 bps	— pass
Fraud proxy (30d)	—	—	+2.2% rel	[+0.3, +4.1] pass
Dispute proxy	—	—	+1.8% rel	120d pending open
Treatment fidelity	99.9%	97.6%	—	floor 95% pass
Credential staleness	—	—	+34% rel	— pass

The vault won on approval and lost on cost. Both are results: 1.4 bps of cost buys 37 bps of approval — 3.8 bps per 100. That exchange rate is what a renewal conversation needs, and we have never had it.

Illustrative figures from the worked example. Credential staleness counts re-authorisation events caused by an expired or reissued credential.

What the vault is worth, per \$1B of qualifying volume

+\$1.16M

a year, to the merchant

+\$11K

to us — interchange-plus

−\$129K

to us — blended pricing

LINE	MERCHANT	US · IC-PLUS	US · BLENDED
Approved volume the vault saves	+\$3.70M	—	—
Merchant gross margin @ 35%	+\$1.30M	—	—
Our net revenue @ 30 bps	—	+\$11K	+\$11K
Vault cost of acceptance @ 1.4 bps	-\$140K	passed through	-\$140K
Net	+\$1.16M	+\$11K	-\$129K

We do not monetize the vault — on blended pricing we fund it. That is a pricing decision, and it is now a decision with a number attached to it.

The part that compounds

DECISION

WHAT HAPPENS

Retire the variant

The merchant-credential arm expires on schedule. The baseline does not move — it was already right

Publish the number

+37 bps [+20, +54] becomes the vault's stated value, with a date and an expiry on the claim

Price the difference

−\$129K per \$1B on blended pricing is now a pricing question with evidence behind it

Re-run on a schedule

Issuer behaviour drifts. A number with no expiry quietly turns back into a claim

Open experiment 2

The same comparison, on clients who do not have the vault yet

The baseline ratchets.

When a variant wins, the baseline moves and next quarter competes with this quarter. When it loses — as this one was meant to — the finding still compounds: the number, the interval and the date enter the evidence record, and the re-run measures drift instead of starting from nothing.

The second experiment is the harder one

Same mechanism, mirrored: control is the merchant’s own credential, variant is the vault network token, and the merchant caps exposure at one or two percent. Everything difficult about it follows from that cap.

MERCHANT’S QUALIFYING VOLUME	AT 2% EXPOSURE	AT 1% EXPOSURE	VOLUME REQUIRED
500K a month · mid-market	~6.5 months	~13 months	3.2M / 6.4M
2M a month · large enterprise	~7 weeks	~14 weeks	3.2M / 6.4M
Pooled across 12 opted-in merchants	~3 weeks	~5 weeks	3.2M / 6.4M

Detecting the same 37 bps needs about 64,900 variant enrolments either way. Halving exposure does not halve the sample — it doubles the volume that must run past it.

Provision ahead, present later

Tokens must be provisioned for the whole eligible population and allowed to season before exposure begins. Provision only the variant at start and we measure new-token performance, not vault performance — and understate ourselves.

Pooling buys speed and costs specificity

A pooled run reaches power in weeks but produces a portfolio number. Each merchant will ask for their own, and at 1–2% most will not have one for months. Decide which we are promising before the first one starts.

Duration is the merchant’s variable, not ours

We control weights everywhere else. Here the cap is contractual, so the honest answer to “how long” is a function of their volume — and it belongs in the conversation before the experiment, not after.

WHY THIS MATTERS BEYOND ONE EXPERIMENT

Every experiment pays twice

Once in the result. Again in the data it leaves behind — which is the data a model needs and the only data that can train one.

1

Run a decision

every choice logs what it picked, what else was permitted, the probability it would pick it, and what the fallback would have done

›

2

Accumulate evidence

that is precisely the shape an off-policy learner trains and validates on — not an approximation of it

›

3

Train a model

the model becomes another strategy class, at the same decision point

›

4

Deploy it the same way

same registration gates, same experiment, same guardrails, same rollback

Without this platform a model has no safe path to production. With it, a model is just another strategy — and the hard part was never the model.

Counterfactual data cannot be created backwards

Propensity and the fallback action can be logged from today's routing and retry engine with no behaviour change — the decision is unchanged, only the record is richer. Every month we do not is a month of traffic that can never train or evaluate anything.

Months of evaluable history available



The gap never closes

A learned strategy needs roughly three months of propensity-logged history before it can be evaluated at all. Delay does not postpone the benefit by the length of the delay — it postpones it by the delay plus the three months, every time.

It changes no behaviour and needs no decision from this room.

WHAT ACTUALLY CHANGES

Before and after, on the things leadership feels

	TODAY	WITH THE PLATFORM
Deterministic change	Module release · weeks to quarters	Registered strategy · days
Evidence	Before-and-after, confounded	Concurrent control, with an interval
Blast radius	Known after rollout	Counted before exposure
Reversal	Redeploy	Change a number
Attribution	"Approval moved"	"The vault, +37 bps, [+20, +54]"
Economics	Approval rate	Approval, cost, fraud, dispute, net
Model deployment	No safe path	Another strategy class
Cumulative value	Unmeasurable	Measured against the holdout

The row that matters most to this programme is the last one. Without a reserved holdout there is no honest answer to "what has all of this been worth".

Three things you should hear from us first

This does not make us faster than Stripe at everything.

Model validation adds weeks to any learned strategy, and we will not shortcut it. What we get for that is strategies defensible to a regulator and to an enterprise procurement team — which is the segment we actually sell to. Deterministic changes are where the days-not-quarters claim holds.

Experiment 1 deliberately runs a weaker arm on live merchant traffic.

Five percent of a vault client's qualifying volume goes through a credential we expect to under-perform. Per \$1B that is \$50M through the weaker arm, roughly \$185K of transactions not approved and about \$65K of merchant margin. Small, bounded and auto-aborting — but real, and visible to the client if they look.

The mechanism does not create the data advantage.

It creates the conditions for one. Thin populations still produce confidently wrong models, and a merchant who opts out of autonomy removes their volume from the loop — and the merchants most able to opt out are the largest.

Five objectives, each with a measure attached

	OBJECTIVE	MEASURED BY	TODAY
01	Collapse the time from hypothesis to production decision	Days from registration to ramp start, by strategy class	Weeks to quarters
02	Make every decision change causally measurable against a fair baseline	Share of decisions attributable to a named strategy, with an interval	Not measured
03	Generate the counterfactual data that models require	Months of propensity-logged history available	Zero
04	Make change safe by construction — bounded, guardrailed, reversible	Guardrail aborts per hundred experiments; rollback time	Rollback = redeploy
05	Make every decision explainable without an engineer	Mean time to explain a decision, self-serve	Escalation

Targets are deliberately absent. A velocity target set before the tooling exists becomes the reason the tooling is skipped.

Fifteen requirements, mapped to objectives

	REQUIREMENT	OBJ	PRI		REQUIREMENT	OBJ	PRI
R1	Strategy is a registered, versioned artefact bound to a decision point	O1	P0	R8	Every decision records propensity and the fallback counterfactual	O3	P0
R2	Flow composition enforces a single difference from its baseline	O2	P0	R9	Live exposure and guardrail monitoring with automatic arm abort	O4	P0
R3	Backtest against logged traffic before any exposure	O1 O4	P0	R10	Per-transaction decision trace, self-serve, under two minutes	O5	P0
R4	Promotion is decoupled from exposure — new flows start at zero	O4	P0	R11	Outcome joins at four horizons, with readouts revised as they land	O2	P0
R5	Experiment registration pre-declares hypothesis, metric, sample size, stopping rule	O2	P0	R12	Readout covers approval, cost, fraud, dispute and net — not approval alone	O2	P0
R6	Assignment is deterministic, at the divergence point, and carried across retries	O2	P0	R13	Promotion ratchets the baseline; rollback is an exposure pointer change	O2 O4	P0
R7	A permanent holdout is reserved and excluded from exploration	O2 O3	P0	R14	Every strategy, flow and experiment carries an owner and an expiry	O4	P1
				R15	Credential provisioning is decoupled from presentation, so treatments season before exposure	O2	P0

R6 and R8 are the two that cannot be retrofitted. R15 is the one most likely to be forgotten — and it is what stops experiment 2 measuring a freshly minted token instead of the vault.

Three things, and only one of them needs a decision

1

Propensity and fallback logging on the current engine

in flight

No behaviour change. The decision is unchanged, only the record is richer. Cannot be backfilled, and every learned strategy depends on it. Should already be in flight.

2

A verified decision-point and capability registry

calendar

Which decision points exist, what each may choose from, and what a route can actually support — verified with processors rather than asserted. Calendar time, not engineering time.

3

A reserved permanent holdout

decision

Two percent, set aside before the first experiment runs. It cannot be created retroactively, and without it we can measure individual experiments but never the platform.

Only item 3 needs a decision today, and it is a decision to reserve two percent of traffic.

THE ASK

Five owners and one reservation

No funding decision today. The mechanism is buildable inside existing platform capacity; what it cannot survive is ambiguity about who owns the five things that bound it.

1	Objective bounds	Who sets how much cost we will accept for approval — and whether that differs by pricing model	Product · Finance
2	Exploration budget	Who owns the basis points we spend learning, and the ceiling on them	Product
3	Decision-point registry	Who decides what becomes an experimentable decision point	Architecture
4	Exposure control	Who may move traffic, and who is accountable when an arm aborts	Platform
5	Exploration consent	Whether a knowingly weaker arm on a named client's live traffic needs contractual consent — and who signs it	Legal · Client Svc

Plus: reserve the two percent holdout this quarter. It is the only thing on this list that gets harder every week we wait.

Should we call this "programmable payment flow"?

What the name gets right

- Flows genuinely are composed, not coded
- It is precise and useful with engineers
- It signals configurability without a release

What it risks

- Implies the merchant writes the logic — they set an objective
- Implies arbitrary code on the payment path, which we forbid by design
- Stripe has owned "programmable" in payments for a decade

Recommendation — keep it internal, lead with the claim that is actually differentiated

Plenty of payment platforms are configurable. Very few can prove that a configuration was better. "Programmable flow" describes the mechanism and is worth keeping in engineering, where it is exact. For leadership and for merchants, the sharper frame is the Decision Platform — every payment decision testable, measurable and reversible. Naming the mechanism sells the thing we share with competitors; naming the evidence sells the thing we would not.